

Comparative Technical Analysis of Image Processing vs Computer Vision

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Course: Computer Vision

Introduction

- **Image Processing** improves the quality of digital images.
- **Computer Vision** helps computers understand images and videos.
- Both use images, but their purpose is different.
- They are used in healthcare, security, and automation.
- This presentation compares their features and applications.

Image Processing – Problem Scope

- Improves the quality of low-quality or noisy images.
- Removes unwanted noise and enhances image clarity.
- Restores damaged or blurred images.
- Compresses images to reduce storage and transmission size.
- Prepares images for further analysis or computer vision tasks.

Computer Vision – Problem Scope

- Detects and identifies objects in images and videos.
- Recognizes faces, text, and different activities.
- Understands scenes and image content automatically.
- Tracks moving objects in real time.
- Supports smart applications like self-driving cars, robots, and surveillance.

Technical Comparison

Image Processing

Enhances or modifies images

Focuses on image quality

Output is an improved image

Uses filters and transformations

Example: Noise removal

Computer Vision

Understands image content

Focuses on image interpretation

Output is information or decisions

Uses AI and Machine Learning

Example: Face recognition

Workflow

Real World Scene



Camera



Image Processing



Feature Extraction



Computer Vision



Decision / Action

Applications

Image Processing

Medical image enhancement

Photo editing

Satellite image processing

Image compression

Document scanning

Computer Vision

Face recognition

Self-driving cars

Object detection

Security surveillance

Industrial quality inspection

Conclusion

- Image Processing focuses on improving and modifying images.
- Computer Vision focuses on understanding and analyzing visual information.
- Both technologies work together to solve real-world problems.
- Image Processing provides better image quality, while Computer Vision enables intelligent decisions.
- Together, they play an important role in modern AI applications.